

Project Planning and Controls

Framework for 100 MW Utility-Scale Solar EPC Operations

1. Department Overview & Philosophy

The Project Planning and Controls (P&C) department serves as the central nervous system of an Engineering, Procurement, and Construction (EPC) project. On a 100 MW utility-scale solar photovoltaic (PV) plant with an associated 33/220kV Grid Substation (GSS), this department ensures that the project lifecycle—from the Limited Notice to Proceed (LNTP) to the Commercial Operation Date (COD)—is executed within the approved time, cost, and quality baselines¹. Beyond merely updating Gantt charts, planning represents a comprehensive discipline encompassing risk management, scope integration, resource allocation, and predictive forecasting. It is the continuous process of aligning the physical reality of the site with the contractual obligations dictated by the FIDIC Yellow Book agreement².

The P&C department sits at the exact intersection of all other project functions. It translates the Engineering department's drawing release dates into the Supply Chain Management (SCM) material requirement dates, which in turn dictate the Construction department's mobilization schedules. It acts as an early warning system, utilizing Earned Value Management (EVM) to detect productivity drops before they threaten the critical path¹. In the context of a simulated EPC environment, the Planning Manager (KRONOS) orchestrates this integration, ensuring that delays in one discipline are dynamically modeled to assess impacts on downstream activities. The Civil & Structural Planning Engineer (VECTOR) and the Electrical & Substation Planning Engineer (WATT) provide discipline-specific granularity, while the MIS & Reporting Engineer (SIGMA) and Cost Control Engineer (LEDGER) synthesize the data into actionable financial and performance intelligence.

Schedule hierarchy is utilized to tailor information density to the appropriate audience, ensuring that executives see strategic milestones while site foremen see daily tasks.

Table_title: Schedule Hierarchy (L1 to L4)

Level	Title	Content and Detail Level	Owner	Primary Audience
L1	Executive Summary Schedule	High-level project phases, major milestones (LNTP, COD), and primary deliverables.	KRONOS	Client Executives, Project Sponsors, C-Suite

		Contains 20-50 activities.		
L2	Summary / Area Schedule	Phased execution by major facility (e.g., PV Field, 33/220kV Substation, Transmission Line). Contains 100-300 activities.	KRONOS	Project Director, Client Project Manager
L3	Control / Baseline Schedule	The fully resource-loaded, logic-driven CPM network detailing engineering packages, procurement MRNs/POs, and construction by discipline. Contains 2,000-5,000 activities.	VECTOR, WATT	Department Heads, Client Engineers, SCM
L4	Execution / Look-Ahead Schedule	Micro-level daily/weekly tasks (e.g., "Pour foundation for Inverter Station 3", "Pull 33kV cable Block A"). Contains	VECTOR, WATT	Construction Managers, Subcontractors, Foremen

		granular constraints and exact crew assignments.		
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2. Schedule Development — Full Detail

The development of the L3 Baseline Schedule for a 100 MW solar project in Uzbekistan requires a meticulous, iterative process spanning from the project kick-off to formal client approval. The process begins with the decomposition of the project scope into a Work Breakdown Structure (WBS), followed by the definition of activities, sequencing, resource estimation, and constraint application³. In the context of a FIDIC Yellow Book contract, the baseline must accurately model the Contractor’s design-build obligations while integrating the Client’s review cycles². The process initiates with KRONOS leading a schedule workshop with engineering, procurement, and construction leads to define the execution strategy, sequentially mapping out the logic from LNTP through engineering design freezes, global procurement, site mobilization, and final commissioning.

The Work Breakdown Structure (WBS) forms the skeletal framework of the project schedule, ensuring every scope item is accounted for without duplication.

Table_title: Detailed WBS for 100 MW Solar PV + 33/220kV GSS

WBS Code	WBS Level Name	Sub-Levels / Deliverables
HEL	HELIOS 100 MW Project	Total Project Roll-up
HEL.10	Milestones	Contractual, Interface, and Payment Milestones
HEL.20	Project Management	Mobilization, EHS, QA/QC, Camp Setup, Handover
HEL.30	Engineering (Design)	Civil, Structural, Electrical (DC/AC), Substation, SCADA
HEL.40	Procurement (SCM)	PV Modules, Inverters, Trackers/MMS, Cables, Main Power Transformer (MPT), Switchgear

HEL.50	Construction - General	Site Clearance, Fencing, Access Roads, Drainage
HEL.60	Construction - PV Field	Piling, MMS Erection, Module Mounting, DC Cabling, Inverter Station Civil/Installation
HEL.70	Construction - Substation	33kV Switchyard, 220kV Switchyard, Control Room Building, MPT Foundation & Erection
HEL.80	Testing & Commissioning	Cold Commissioning, Hot Commissioning (IEC 62446), Grid Synchronization, PR Test

The critical path for a 100 MW solar EPC project rarely runs through the PV field construction, which is highly modular, scalable, and float-heavy. Instead, the true critical path invariably flows through the Engineering, Procurement, and Construction of the 220kV Grid Substation. The longest continuous sequence typically dictates: LNTP Award → Substation Electrical Design → Client Approval → Manufacturing and Factory Acceptance Testing (FAT) of the 220kV Main Power Transformer (MPT) (requiring 10–12 months) → Logistics and Customs clearances, specifically at bottlenecks like the Khorgos border (1–2 months)⁴ → Transformer Erection → Pre-commissioning → Grid Synchronization with the National Electric Grid of Uzbekistan (NEGU)⁶. A secondary, near-critical path frequently involves the procurement of PV modules, which is heavily influenced by global supply chain volatility and raw material availability⁸.

Table_title: Standard Milestone List for Solar EPC

Milestone	Description	Typical Timing
LNTP / NTP	Limited / Full Notice to Proceed.	Day 0

Engineering Freeze	Approval of basic design and major equipment sizing.	Month 3
Site Mobilization	Establishment of site camp and preliminary works.	Month 2
First Pile Driven	Commencement of PV structural foundations.	Month 4
First Module Delivered	Arrival of initial PV module batches at the site.	Month 6
Mechanical Completion	All physical works complete; ready for pre-commissioning.	Month 15
First Energization (Back-feed)	Drawing power from NEGU grid to test substation equipment.	Month 16
Grid Sync (COD / PAC)	First export of power / Provisional Acceptance Certificate.	Month 18
FAC	Final Acceptance Certificate (End of Defect Liability Period).	Month 42

In Primavera P6, the schedule is constructed using specific conventions to ensure scalability. Activity codes are globally or project-defined parameters (e.g., Discipline, Area, Subcontractor, Phase) utilized to filter and group tasks independently of the rigid WBS structure³. Calendars are highly critical in the Central Asian context. The baseline must utilize custom calendars accounting for severe winter constraints, such as frozen ground preventing piling, and sub-zero temperatures requiring specialized thermal blankets for concrete curing in accordance with SHNK national building regulations¹⁰. Typically, civil activities utilize a 6-day weather-constrained calendar that accounts for winter productivity losses, while engineering utilizes a 5-day standard calendar. Procurement lead times are mapped dynamically into the network logic. The schedule links the

"Approved for Construction" (AFC) drawing milestone directly to the Material Requisition Note (MRN) submission. This MRN drives the Purchase Order (PO) issuance, which links to the manufacturing duration, FAT, shipping, customs clearance, and finally, site delivery. The delivery activity then acts as a Finish-to-Start (FS) predecessor to the physical installation activity on site.

The most common scheduling mistakes in solar EPC include excessive use of mandatory constraints (which artificially suppress negative float and mask the true critical path), failing to account for 14-to-21-day client review cycles in engineering, neglecting logistics bottlenecks, and assuming uniform labor productivity across extreme summer and winter seasons³.

Furthermore, planners frequently fail to adequately detail the commissioning phase, treating it as a single block rather than a complex sequence of IEC 62446 Category 1 and Category 2 tests¹³.

3. Daily, Weekly, Monthly Planning Workflow

The execution phase demands a relentless rhythm of data collection, analysis, and reporting to maintain control over the EPC lifecycle. On a 100 MW site in Uzbekistan, a typical day for the Planning Manager (KRONOS) begins with a 07:30 AM site walk alongside the construction managers to visually verify the previous day's progress and identify undeclared roadblocks. At 08:30 AM, KRONOS leads the daily coordination meeting. The Civil Planning Engineer (VECTOR) and the Electrical Planning Engineer (WATT) spend their mornings engaging with subcontractors and site supervisors to collect exact quantities, such as the number of piles driven, trenches excavated, or modules mounted. By the afternoon, the MIS & Reporting Engineer (SIGMA) aggregates these inputs, resolving discrepancies between QA/QC inspection logs and subcontractor claims, ultimately generating the Daily Progress Report (DPR).

Schedule updates in Primavera P6 are formally triggered at the end of every week for internal review, and at the end of every month for official client submission. The 3-week look-ahead (3WLA) is the primary tactical tool used in weekly site meetings. It is built by extracting the upcoming 21 days of activities from the L3 baseline, filtering them by discipline and area. VECTOR and WATT demand that site engineers assign specific crews, equipment, and material allocations to each task. The 3WLA is only approved if all constraints—such as drawing approvals, material availability, and safety permits—are cleared for the first week of the look-ahead period.

The reporting architecture relies on a strict roll-up methodology. The Daily Progress Report (DPR) captures the raw daily inputs. It contains the weather conditions (crucial for supporting future EOT claims based on adverse weather), manpower histograms, equipment logs, and daily quantities installed. It is compiled by SIGMA and submitted by 18:00 hours daily. The Weekly Progress Report (WPR) rolls up the daily quantities into a holistic view. It contains the 3-week look-ahead, critical path analysis, procurement status tracking (including manufacturing progress and shipping ETAs), and an updated progress S-curve. The Monthly Progress Report (MPR) is the definitive contractual document. It contains an executive summary, detailed schedule narrative, formal EVM calculations (SPI, CPI), risk register updates, delay event logs, quality/HSE statistics, payment milestone status, and comprehensive

photographic evidence. The MPR is authored collaboratively by all department heads but compiled and finalized by KRONOS before Project Director approval.

The S-curve is the visual representation of cumulative progress over time, serving as the most recognizable metric of project health. It is built by plotting the baseline planned value (BCWS) against the time axis, resulting in an "S" shape due to slow initial mobilization, rapid construction peaks, and tapered commissioning. Each reporting period, the actual physical progress (BCWP) is plotted against the baseline. If the actual line falls below the planned line, the project is delayed.

The schedule narrative accompanies the MPR, providing a qualitative explanation of the quantitative data. It is written by KRONOS at the end of the month and must cover the current progress percentage, a detailed explanation of any variances between planned and actual progress, an analysis of the current critical path, a summary of delays encountered, and the specific mitigation strategies implemented by the contractor to recover lost time.

4. Progress Measurement & Earned Value

Physical progress measurement on a solar EPC project requires a robust, objective, and pre-approved Weighting System. Progress cannot be claimed subjectively based on an engineer's visual estimate; it must be tied to measurable, physical installation.

Table_title: Typical EPC Level 1 & Level 2 Weighting System

Major Phase / Discipline	L1 Weightage (%)	L2 Sub-Weightage (%)
Engineering	5%	Basic Design (2%), Detailed Design (3%)
Procurement	60%	Modules (30%), MPT (10%), Trackers (10%), BOS (10%)
Construction	30%	Civil (10%), Mechanical (10%), Electrical (10%)
Testing & Commissioning	5%	Cold Comm (2%), Hot Comm (2%), Grid Sync/PR (1%)

Within the construction phase, progress is further subdivided into specific activities and tracked using stringent Rules of Credit. For civil works, concrete is tracked by cubic meters (m^3) poured, but the credit is staggered to prevent front-loading (e.g., 20% for excavation, 30% for formwork and rebar, 40% for the pour, and 10% for curing and stripping). Structural erection is tracked by tonnage or the number of MMS tables fully bolted and torqued. Electrical

works, such as DC cabling, are tracked by linear meters, with credit split between trenching, cable laying, backfilling, and final insulation resistance testing (Megger)¹⁴.

Earned Value Management (EVM) translates this physical progress into financial or man-hour metrics to objectively assess performance¹. The core EVM metrics calculated by KRONOS and LEDGER include:

- **BCWS (Budgeted Cost of Work Scheduled) / Planned Value (PV):** The authorized budget assigned to the work to be accomplished up to a specific date.
- **BCWP (Budgeted Cost of Work Performed) / Earned Value (EV):** The measure of work performed expressed in terms of the budget authorized for that specific work.
- **ACWP (Actual Cost of Work Performed) / Actual Cost (AC):** The realized cost incurred for the work performed to date.
- **SPI (Schedule Performance Index):** Calculated as $SPI = \frac{BCWP}{BCWS}$. An $SPI < 1.0$ indicates the project is behind schedule.
- **CPI (Cost Performance Index):** Calculated as $CPI = \frac{BCWP}{ACWP}$. A $CPI < 1.0$ indicates the project is over budget.

For example, if the L3 baseline mandated the installation of 50,000 PV modules by the end of Month 8, at a budgeted installation cost of \$50 per module, the BCWS is \$2,500,000. If the site team only manages to install 40,000 modules by that date, the BCWP is \$2,000,000. The SPI is 0.8, signaling a severe 20% schedule deficit for that activity.

The planning department interfaces heavily with the Billing and Quantity Surveying (QS) departments to validate earned value against payment milestones. The physical progress approved in the P6 schedule by KRONOS must exactly match the Interim Payment Certificates (IPC) submitted to the client. If SIGMA reports that only 40,000 modules are installed, the QS department cannot bill the client for 50,000 modules, ensuring strict financial governance.

5. Delay Management & EOT

Delay management requires proactive, prospective identification and rigorous documentation. A planning engineer identifies a delay by analyzing the L3 schedule update and noting negative float on the critical or near-critical paths. Triggers include unmet predecessor dependencies (such as delayed NEGU grid connection technical approvals⁶), severe resource shortages, or sudden force majeure events.

The Delay Event Log is maintained meticulously by KRONOS, recording the event ID, date of occurrence, detailed description, affected activities, party responsible, and the contractual notice dates. Under FIDIC Yellow Book Sub-Clause 20.1 (1999 edition) or Clause 20.2 (2017 edition), the Contractor must issue a formal Notice of Claim within 28 days of becoming aware of the delaying event, failing which the right to an Extension of Time (EOT) and additional payment is permanently forfeited².

Delays in EPC contracting are categorized into three primary types:

1. **Excusable and Compensable:** Employer risk events, such as late handover of the site, changes in client scope, or delayed approval of engineering drawings beyond the contractual review period. The contractor is entitled to an EOT and prolongation costs².

2. **Excusable but Non-Compensable:** Force majeure events, such as unprecedented extreme winter storms in Uzbekistan completely halting civil works. The contractor is entitled to an EOT to avoid Delay Damages (Liquidated Damages), but must generally bear their own site overhead and prolongation costs.
3. **Culpable (Non-Excusable):** Contractor risk events, such as subcontractor insolvency, internal procurement delays, or failure to mobilize adequate resources. No EOT or costs are granted, and the contractor faces punitive Delay Damages if the COD is missed².

To support an EOT claim, contemporaneous records are absolutely essential. These include signed DPRs, certified weather station data, formal correspondence, Requests for Information (RFIs), and dated, geotagged photographic evidence.

The Society of Construction Law (SCL) Delay and Disruption Protocol recommends the **Time Impact Analysis (TIA)** as the preferred methodology for prospective delay analysis¹⁵. The preparation of a TIA involves specific sequential steps:

1. Identifying the delay event and calculating its exact duration.
2. Updating the baseline schedule to the moment just before the delay occurred, creating the "unimpacted" schedule.
3. Inserting the delay event as a new activity (or fragnet) linked to the specific affected sequence.
4. Recalculating the schedule logic.
5. The difference in the overall project completion date between the unimpacted and impacted schedules demonstrates the contractor's entitlement to an EOT.

Other retrospective methods include Windows Analysis (slicing the project into discrete time periods to assess delays as they evolved), As-Planned vs As-Built (a basic visual comparison, often rejected in formal arbitration for complex EPCs due to its simplicity), and Collapsed As-Built (removing delays from the final as-built schedule to prove when the project would have finished without employer interference). The planning team works symbiotically with the Contracts team; KRONOS provides the causation link and delay quantum via the TIA, while the Contracts Manager frames the legal and contractual entitlement under FIDIC Sub-Clause 8.4².

6. Cost Control (LEDGER Agent)

Cost Control on a solar EPC project bridges the gap between schedule progress and financial health. The Cost Control Engineer (LEDGER) establishes the project budget by mapping it directly to the WBS and establishing a Cost Breakdown Structure (CBS). This ensures that every dollar spent can be traced back to a specific physical activity in the schedule.

Table_title: Standard EPC Cost Codes for Solar Projects

Cost Code Category	Description of Scope
DIR.CIV.FND	Direct Cost - Civil - PV Foundations (Piling, Concrete)
DIR.MEC.MMS	Direct Cost - Mechanical - Module

	Mounting Structures
DIR.ELE.DC	Direct Cost - Electrical - DC Cabling, Combiner Boxes
DIR.SUB.HV	Direct Cost - Substation - High Voltage Equipment (MPT, GIS)
IND.ENG.DES	Indirect Cost - Engineering and Design Subcontracts
IND.SCM.FRG	Indirect Cost - Procurement - Freight, Logistics, Customs
IND.SIT.OVH	Indirect Cost - Site Overheads (Camp, EHS, Management Salaries)

The Cost Report is produced monthly by LEDGER and reviewed by the Project Director and corporate finance. It tracks the original tender budget, approved variations, the current working budget, financial commitments (issued POs and signed subcontracts), costs incurred to date, the Estimate to Complete (ETC), and the Estimate at Completion (EAC).

The Budget Variance Report highlights negative deviations from the baseline. Escalation to executive management is triggered when the Cost Performance Index (CPI) drops below 0.95 or when specific cost codes exceed a predefined contingency threshold. Earned value directly feeds the cost control function; the Earned Value (BCWP) provides an objective, schedule-linked measure of what the completed physical work *should* have cost. This allows LEDGER to identify if subcontractors are overbilling relative to their physical output, or if bulk material consumption is exceeding the design quantities.

Cash flow forecasting relies entirely on the P6 schedule. By extracting the resource-loaded costs from the schedule over the project timeline, LEDGER generates the planned cash-out S-curve. This cash-out profile is then overlaid against the expected contractual payment milestones (cash-in) to predict working capital requirements, ensuring the EPC contractor remains cash-flow positive and can fund the massive early procurement requirements for PV modules and the MPT.

7. MIS & Reporting (SIGMA Agent)

The Management Information Systems (MIS) and Reporting Engineer (SIGMA) acts as the data custodian for the project, relying on a sophisticated technology stack encompassing Primavera P6, Microsoft Excel (utilizing Power Query and VBA macros), Power BI for executive dashboards, and field data collection applications (e.g., Fieldwire, Dalux) used by site

inspectors.

The roll-up of data from the field to the boardroom demands rigorous reconciliation checks. SIGMA ensures that the sum of the daily module installations recorded in the DPRs exactly matches the weekly total in the WPR, which must subsequently match the monthly MPR and the physical update entered into P6. If subcontractor A claims 5,000 meters of cable trenched on their invoice, but the QA/QC inspection logs only show 4,200 meters passed testing, SIGMA caps the progress at 4,200 meters, enforcing the rule that progress is only earned upon quality acceptance.

Table_title: Planning MIS Dashboard KPIs

KPI Metric	Description	Warning / Escalation Trigger
Schedule Performance Index (SPI)	Earned Value / Planned Value	< 0.95 (Amber status), < 0.90 (Red status)
Critical Path Float	Total Float remaining on the longest path	< 15 days
Milestone Hit Rate	Percentage of baseline milestones achieved on time	$< 80\%$
RFI Turnaround Time	Average days for client/engineer to resolve RFIs	> 7 days
Design Approval Status	Percentage of drawings Approved for Construction	Delays > 14 days past baseline dates

Standard MPR visualizations generated by SIGMA include the Progress S-Curve (plotting Planned vs Actual vs Forecast), manpower histograms (showing planned vs actual labor peaks to highlight resource shortages), commodity installation bar charts (e.g., modules installed per week vs the required target run-rate), and 90-day look-ahead Gantt charts. The reporting engineer catches inconsistencies by cross-referencing independent data sources—for example, comparing warehouse material issuance logs generated by SCM against subcontractor installation claims to prevent fraudulent front-loading of progress.

8. Interfaces with Other Departments

The Planning department does not exist in a vacuum; it is a highly dependent function that integrates constraints from every other discipline to maintain a realistic schedule.

- **With Engineering:** The Master Document Register (MDR) drawing release schedule is linked directly to the P6 construction schedule. If structural drawings for the 220kV gantry are delayed, the civil works are pushed back, consuming float. Planning models these delays to alert management before construction crews are mobilized and left idle.
- **With SCM:** The procurement schedule tracks the Material Requirement List (MRL). KRONOS reverse-engineers the required on-site date from the construction schedule, subtracts logistics, manufacturing, and internal procurement lead times to give SCM the precise "Drop Dead Date" for PO issuance.
- **With Construction:** The data flow is constant and bidirectional. VECTOR and WATT gather daily physical quantities from site supervisors, validate these against QA/QC releases, and feed the upcoming 3WLA back to the construction teams to dictate the upcoming sequence of work.
- **With Quality (QA/QC):** Inspection Hold Points and Witness Points (ITPs) act as hard constraints in the schedule. Crucially, testing regimes such as the IEC 62446 commissioning tests (insulation resistance, string open-circuit voltage, IV curve tracing, thermography) must be sequenced logically at the end of the project¹³. Testing cannot occur until specific mechanical and electrical predecessors are complete, and test failures necessitate rework loops which rapidly consume schedule float.
- **With EHS:** Safety incidents (e.g., a near-miss halting work in a specific block pending investigation) or Permit to Work (PTW) delays are captured in the schedule as non-working days in the calendar, providing a documented explanation for localized productivity drops.
- **With Contracts:** The planning team identifies the critical path impact of variations and generates the Time Impact Analysis (TIA) for Extension of Time (EOT) claims. This technical schedule evidence enables the Contracts department to formally notify the client under FIDIC rules and negotiate prolongation costs².
- **With Client:** The client's planning representative heavily scrutinizes the baseline and monthly updates. Common client objections include unsupported changes in baseline logic, the artificial suppression of float using hard constraints (like "Mandatory Finish"), the front-loading of early payment milestones, and optimistic acceleration logic that fails to show a corresponding increase in resource loading.

9. Real Document Templates — Full Detail

A. Daily Progress Report (DPR) Template

Table_title: DPR Template Structure and Populated Example

Section	Fields & Realistic Data (HELIOS 100 MW Project)
1. Header Information	Date: 12-Nov-2026 Project: HELIOS

	100MW Weather: Max 12°C, Min -2°C (Clear)
2. HSE Statistics	LTI: 0 Near Misses: 1 Tool Box Talks Conducted: 4 PTW Issued: 15
3. Manpower & Equipment	EPC Staff: 45 SubC Civil: 120 SubC Mech: 80 Excavators: 6 Piling Rigs: 4
4. Civil Progress (Daily/Cum)	Site Clearing (m^2): 0 / 1,000,000 Piles Driven (Nos): 300 / 15,400 Inverter Fnd Concrete ($\$m$): 45 / 450
5. Mech/Elec Progress	MMS Tables Erected: 15 / 850 Modules Installed: 600 / 181,818 DC Cable (m): 2,500 / 120,000
6. Substation Progress	33kV Switchgear Erection: Ongoing MPT Bay Civil: Formwork in progress
7. Top Issues/Blockers	1. SubC A lacking sufficient piling rigs. 2. Frozen ground in Block 4 slowing trenching.

B. Weekly Progress Report (WPR) Template

Table_title: WPR Template Structure

Section	Content Description & Realistic Example
1. Executive Summary	"Week 24: Overall progress stands at 32.4% against a planned 35.1%. Delay is primarily driven by MPT FAT delays."
2. Schedule & EVM	SPI: 0.92. CPI: 0.98. Critical path remains firmly on 220kV GSS civil works.
3. Engineering Status	MDR Status: 85% AFC. Pending: SCADA architecture approval from NEGU.

4. Procurement Status	MPT manufacturing complete; FAT scheduled for next Tuesday. Tracker steel delayed at Khorgos border.
5. Look-Ahead (3WLA)	Next 21 Days: Complete Block 1-4 piling; Commence Block 1 module erection; Complete MPT oil pit concrete pour.

C. Monthly Progress Report (MPR) Template

Table_title: MPR Section-by-Section Structure

Section	Owner/Author	Content Definition
1. Project Overview	Project Director	High-level summary, commercial status, executive highlights.
2. HSE & Quality	EHS / QA Managers	Lagging/leading indicators, NCRs raised/closed, IEC 62446 test results status ¹⁴ .
3. Schedule Narrative	Planning Manager	Explanation of schedule variances, critical path analysis, mitigation measures.
4. Progress S-Curves	SIGMA	Visual graphs plotting EVM data (Planned vs. Actual vs. Forecast).
5. Design & SCM	Eng / SCM Managers	MDR progress table, MRL tracker, manufacturing updates, shipping logs.
6. Construction	Construction Mgr	Detailed physical progress by discipline with photographic evidence.

7. Risk & Delay Log	P&C / Contracts	Updated risk register, TIA summary, EOT notices logged.
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D. 3-Week Look-Ahead (3WLA) Schedule Format

Table_title: 3WLA Data Structure

Act ID	Activity Name	Total Float	Remaining Duration	Wk 1 Target	Wk 2 Target	Wk 3 Target	Constraints/Blockers
HEL-C-10	Pile Driving Blk 5	10 d	15 d	1,000 piles	800 piles	-	Awaiting steel delivery
HEL-E-45	33kV Trenching	5 d	20 d	2,000 m	2,500 m	2,500 m	Ground freezing

E. S-Curve Backend Data Structure

Table_title: S-Curve Backend Database

Month	BCWS (Period)	BCWS (Cumulative %)	BCWP (Period)	BCWP (Cumulative %)	Variance (%)
M1 (Jan)	\$1.0M	2.0%	\$0.9M	1.8%	-0.2%
M2 (Feb)	\$2.5M	7.0%	\$2.0M	5.8%	-1.2%

F. Schedule Narrative (Realistic Example Text)

"During the month of November, overall progress reached 42.1%, trailing the baseline target of 45.3%. The Schedule Performance Index (SPI) currently stands at 0.93. The primary variance is driven by a 14-day delay in the delivery of the 220kV Main Power Transformer due to customs clearance bottlenecks at the Khorgos border⁴. This event has consumed the remaining 10 days of project float, pushing the critical path directly through Substation Electrical Erection. In the PV field, piling operations in Blocks 4 and 5 encountered heavily frozen ground due to early winter conditions, slowing ramming productivity by 25%. To mitigate this, the contractor has

deployed two additional hydraulic piling rigs and shifted night-shift resources to DC cabling where thermal impact is lower. The 28-day notice for the border delay was submitted under FIDIC Sub-Clause 20.1 on Nov 15th²."

G. Delay Event Log Format

Table_title: Delay Event Log

ID	Event Description	Date Occurred	Notice of Claim Date	Affected Path	Estimated Delay	Status
D-04	NEGU Grid Design Change	10-Oct	15-Oct	GSS Engineering	21 Days	TIA Submitted
D-05	Severe Snowstorm	02-Nov	05-Nov	Civil Works	4 Days	EOT Approved

H. Schedule Baseline Approval Memo

Contains: Project Name (HELIOS 100 MW), Revision Number (Rev 0), Data Date, Total Number of Activities (3,450), Critical Path description (Substation GSS sequence), Total Project Value, Signatures of Contractor Project Director, Planning Manager (KRONOS), and Client Representative. It includes explicit assumptions regarding weather calendars, site access handover dates, and client review durations.

10. Software & Tools

Primavera P6 serves as the undisputed industry standard for complex EPC scheduling³. Activity IDs follow a strict naming convention to enable database sorting and the conversion of WBS structures to activity codes for advanced filtering¹⁹. For example, an ID formatted as HEL-CIV-PV-STR-1050 equates to Project (HELIOS) - Discipline (Civil) - Area (PV Field) - Component (Structures) - Sequence (1050).

Calendars are configured meticulously to reflect real-world constraints. In Uzbekistan, a solar EPC requires at least three distinct calendars: a 7-day calendar for automated manufacturing and shipping; a 5-day calendar with public holidays for Engineering and Client Reviews; and a 6-day weather-constrained calendar for Site Construction. This construction calendar specifically reduces working hours or blocks out days during the harsh winter months (December to February) to account for snow delays and the SHNK thermal insulation requirements for concrete pouring in sub-zero temperatures¹⁰.

Relationships in solar EPC predominantly utilize standard Finish-to-Start (FS) links, but fast-tracking requires Start-to-Start (SS) links with lags. For instance, Trenching (SS+5 days)

→ Cable Laying ensures pipe laying begins 5 days after trenching starts, allowing concurrent work fronts. Constraints in P6 must be used sparingly. Soft constraints (like "Start On or After") are acceptable to model actual material arrival dates. However, hard constraints (like "Mandatory Start" or "Mandatory Finish") override schedule logic and are highly frowned upon by client planners, as they artificially lock the schedule and hide true critical path float³. For client submissions, SIGMA exports the standard Gantt Chart, a Critical Path Filter layout, and tabular reports detailing Early Start/Finish dates and Total Float. Alongside P6, Excel is heavily used for daily quantity tracking and complex data manipulation, while Power BI is deployed for executive dashboards, connecting directly to P6 XML exports and Excel databases to visualize the S-Curves and histograms interactively.

11. Common Problems & How Planning Handles Them

Problem 1: Module delivery delayed by 6 weeks.

Planning Response: KRONOS models the 6-week delay in P6. If this pushes the mechanical completion past the float threshold, planning implements mitigation logic known as out-of-sequence work. VECTOR directs the civil teams to bypass module mounting in the affected blocks and instead accelerate the completion of all civil and DC cabling works. When the modules finally arrive, WATT floods the site with mechanical erection crews, working overlapping shifts to crash the schedule and recover the lost time.

Problem 2: Design drawings not released on time. *Planning Response:* SIGMA tracks the client review cycle in the MDR. If the 14-day contractual review period under the FIDIC Yellow Book is breached, KRONOS immediately enters a delay event in the log¹². A Time Impact Analysis (TIA) is prepared showing that the delay in the Substation foundation drawing is directly driving the critical path. This provides the Contracts department with the foundation for a compensable EOT claim.

Problem 3: Monsoon / extreme winter weather in Central Asia causing site shutdown.

Planning Response: Uzbekistan winters can freeze the ground, making piling impossible and halting concrete works²⁰. The baseline calendar should have anticipated normal winter slowdowns based on historical meteorological data. If the weather is exceptionally severe (e.g., a 1-in-50-year storm), it constitutes Force Majeure. Planning generates weather downtime reports cross-referenced with local meteorological data, submits a delay notice, and re-sequences indoor work (e.g., control room wiring) to minimize labor standing time.

Problem 4: Subcontractor productivity lower than baseline.

Planning Response: EVM metrics highlight this failure immediately (e.g., the SPI drops to 0.7 for cabling). Planning runs a resource analysis showing the subcontractor is using 40 men instead of the planned 80. The Project Manager uses this objective P6 data to invoke contractual default clauses, forcing the subcontractor to augment manpower immediately or allowing the EPC to bring in a supplementary subcontractor and back-charge the costs.

Problem 5: Client issues a scope change mid-project (e.g., adding a BESS reserve).

Planning Response: The client requests space and grid connection for a future Battery Energy Storage System (BESS)⁶. KRONOS creates a separate "fragnet" (a fragment of a network schedule) detailing the engineering, procurement, and civil works required for this addition.

This fragnet is inserted into the baseline. If the added logic pushes the COD, a formal variation order encompassing both time (EOT) and cost (prolongation and direct costs) is formulated by LEDGER and KRONOS, and submitted to the client for approval before executing the new work.

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